Pharmacological and Psychological Effects of Caffeine Ingestion in 40-km Cycling Performance

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ABSTRACT

FOAD, A. J., C. J. BEEDIE, and D. A. COLEMAN. Pharmacological and Psychological Effects of Caffeine Ingestion in 40-km Cycling Performance. *Med. Sci. Sports Exerc.*, Vol. 40, No. 1, pp. 158–165, 2008. The ergogenic effects of caffeine are well documented. Research has yet to examine any psychological contribution to this effect. **Aim:** To explore the psychological and pharmacological effects of caffeine in laboratory cycling performance. **Method:** Fourteen male competitive cyclists performed 14 40-km time trials (eight experimental interspersed with six baseline). The experimental phase consisted of two trials for each of four experimental conditions: informed caffeine/received caffeine, informed no treatment/received caffeine, informed caffeine/received placebo, and informed no treatment/received no treatment. Conditions were nonrandomized. ANOVA was used to estimate main effects and interactions for mean values of power, heart rate, blood lactate, and maximal oxygen uptake. Probabilistic inferences for mean power were based on a smallest worthwhile change of 1.5%. **Results:** Relative to baseline, a very likely beneficial main effect of receiving caffeine (3.5%; 95% confidence interval 1.5 to 5.5%), and a possibly beneficial main effect of being informed of caffeine (0.7%; −0.7 to 2.1%) were observed. A substantial interaction between belief and pharmacology indicated that caffeine exerted a greater effect on performance in conditions when subjects were informed that they had not ingested it, whereas belief exerted a greater influence on performance in the absence of caffeine (2.6%; −0.7 to 5.9%). A possibly harmful negative placebo (nocebo) effect was observed when subjects were correctly informed that they had ingested no caffeine (−1.9%; −4.1 to 0.3%). No clinically significant changes relative to baseline were observed in mean heart rate. Clear and substantial increases in blood lactate were evident after receipt of caffeine. Data for mean oxygen uptake were unclear. **Conclusion:** Our data support the ergogenic efficacy of caffeine but suggest that both positive and negative expectations impact performance. **Key Words:** PLACEBO EFFECT, BALANCED PLACEBO DESIGN, BELIEF EFFECTS, ERGOGENIC AIDS, PLACEBO RESPONSIVENESS

The ergogenic effects on sports performance of a wide range of substances are well documented. It has been suggested, and demonstrated empirically (7,10), that whereas these substances exert a "true" pharmacological or nutritional effect, a percentage of their overall ergogenicity might be attributable to the placebo effect: a positive outcome driven by the belief or expectation that a beneficial treatment has been received. Several experimental studies in sport have demonstrated significant placebo effects on performance (2,6,7,10,21). Furthermore, a large body of research elsewhere has demonstrated such effects on variables such as pain, inflammation, immune function, mental health, and learning (for a review, see Evans (13)).

Although some idea of the magnitude of the placebo effect on sports performance is emerging (6,7,10), several questions remain unaddressed. One such question relates to the degree to which placebo effects interact with physiological effects; do they share the same headroom for improvement, or are they simply additive? If the former is true, this should be of concern to researchers using experimental designs in which the effect size is calculated as the difference between placebo control and experimental conditions. If the latter is correct, nutritionists and sports science consultants would theoretically be able to augment the physiological effects of ergogenic substances by manipulating athletes' beliefs about those substances.

The question above is, however, problematic to address, especially because the placebo-controlled design often employed in sports research does not provide sufficient information. That is, although the effect of two or more conditions on the dependent variable are observed, the

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subjects' belief about which condition they have been allocated to is held constant by blinding. Also, largely because of legitimate resource or experimental constraints, experimental and placebo conditions are rarely augmented by baseline or nonplacebo controls, rendering estimates of effect sizes over true baseline impossible. This latter limitation is important, given recent research suggesting that the magnitude of placebo effects might be determined by a dose–response relationship, with stronger beliefs associated with larger effects (7,20), and negative beliefs potentially exerting a negative effect on performance (6). It is evident that to explore the placebo effect, the traditional, placebo-controlled trial is inadequate.

The present study used the balanced placebo design (22) in an attempt to examine the placebo effects of caffeine ingestion in 40-km laboratory cycling performance. This design provides a powerful method for clarifying the relative effects of belief and pharmacology as well as their interactions, and it has the potential to yield information that could not be obtained using a traditional, double-blind trial (20). Such information includes whether the pharmacological and placebo effects of caffeine ingestion are additive or interactive, whether any placebo component of caffeine ingestion is equivalent to the placebo effect in the traditional placebo control condition, and the magnitude of the pharmacological effect of caffeine independent of the placebo response. In the present study, response to the treatment was operationalized in terms of volitional power output in cycling performance.

METHOD

Design. The present study had full institutional ethics approval, and all subjects read and signed informed consent documentation before data collection. The aim of the study was to use a 2 (drug: caffeine/no caffeine) × 2 (belief: caffeine/no caffeine) within-subjects repeated-measures design to quantify the placebo and pharmacological effects of caffeine on 40-km cycling performance. This approach required the subjects to be deceived about the substance they had ingested in both the informed caffeine/received placebo (i.e., placebo) and informed no treatment/received caffeine (i.e., antiplacebo) conditions. A comprehensive debrief process, as required by our institutional ethics committee and by the guidelines of the American Psychological Association (1), was carried out immediately after data collection. The study was conducted on a single-blind basis.

Subjects. Subjects were 14 male road cyclists (age 43 ± 7 yr; height 181 ± 9 cm; mass 78.1 ± 6.3 kg; daily coffee consumption 310 ± 75 $mg \cdot d^{-1}$; $\dot{V}O_{2max}$ 51.7 ± 1.1 $mL \cdot kg^{-1} \cdot min^{-1}$; MMP 398 ± 44 W). All subjects were competing regularly at the time of the study. Habitual caffeine consumption (derived from Maughan (24)) and general health status were ascertained via prestudy questionnaire. To ensure familiarity with the effects of caffeine and to control for individual differences in reactivity to caffeine from caffeine habitation, only moderate caffeine users (ingesting approximately 300 $mg \cdot d^{-1}$) were included in the study. Although no difference in placebo response by gender has been reported (3), males were recruited to avoid confounding the experimental manipulation by a reduced rate of caffeine clearance, both during the late luteal phase of the menstrual cycle and among oral contraceptive users. In the informed consent documentation, subjects were informed that the aim of the study was to assess the effects of caffeine ingestion on cycling performance. They were also informed, in accordance with the ethical guidelines of the American Psychological Association (1), that they could receive either caffeine or a placebo in any one trial. However, they were not informed about the true purpose of the study or that they would be deceived in the informed caffeine/received placebo and informed no treatment/received caffeine conditions.

Statistical power and smallest worthwhile effects. Sample size estimations were conducted before data collection. Fewer than 10 subjects were estimated to provide an *a priori* statistical power of 0.80. This estimate was based on an α-value of 0.05, a low CV for power (1.5%) derived through pilot work before the present study, and the smallest practically beneficial improvement in performance estimated as an approximate 1.5% increase in power output (25). (Given that in the present study, cycling performance was used as a model system for the possible effects of placebo and caffeine, and, furthermore, that effects below 1.5% could be of significance to athletes, we have also reported effects using smallest worthwhile effects of 1.0% and 0.5% in Tables 1 and 2 below.) Fourteen subjects were recruited to allow for possible participant attrition, although all subjects subsequently completed the study.

TABLE 1. Factorial ANOVA of the effects of caffeine ingestion and belief of caffeine ingestion on performance and chances that effects are beneficial/trivial/harmful.

	Difference (%) in Mean Power (W)		Chances Effect Is Beneficial (Trivial/Harmful) for Smallest Worthwhile Effect of…		
Condition	Mean ± SD	95% CI	0.5%	1.0%	1.5%
Caffeine ingestion[a]	3.5 ± 3.4*	1.5 to 5.5	100 (0/0)	99 (1/0)	98 (2/0)
Belief of caffeine ingestion[b]	0.7 ± 1.6	−0.7 to 2.1	62 (34/5)	33 (66/1)	12 (88/0)

* Significant at $P < 0.01$.

95% CI, 95% confidence interval.

[a] ((Inform caffeine, give caffeine + Inform no caffeine, give caffeine) − (Inform caffeine, give placebo + Inform no treatment, give no treatment))/2.

[b] ((Inform caffeine, give caffeine + Inform caffeine, give placebo) − (Inform no treatment, give caffeine + Inform no treatment, give no treatment))/2.

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TABLE 2. Effects of experimental conditions on mean power output.

Condition	Power (W)	Change in Power (%) from Mean Baseline		Chances Effect Is Beneficial (Trivial/Harmful) for Smallest Worthwhile Effect of…		
		Mean ± SD	95% CI	0.5%	1.0%	1.5%
Inform caffeine/give caffeine	277 ± 25	2.3 ± 3.3*	0.3 to 4.3	96 (3/1)	92 (8/0)	82 (18/0)
Inform no treatment/give caffeine	279 ± 26	2.9 ± 3.4**	1.3 to 4.5	99 (0/0)	98 (1/0)	96 (4/0)
Inform caffeine/give placebo	272 ± 27	0.1 ± 3.2	−4.0 to 4.2	41 (23/36)	30 (44/26)	21 (60/18)
Inform no treatment/give no treatment	266 ± 30	−1.9 ± 3.8	−4.1 to 0.3	2 (6/92)	1 (16/83)	1 (33/67)

* $P < 0.05$; ** $P < 0.01$.
95% CI, 95% confidence interval.

Performance measure. Average power during a self-paced 40-km time trial was employed as the performance measure. This protocol has been demonstrated as reliable (17) and is representative of road cycling time-trial competition (27).

Equipment. Each 40-km time trial was completed on the SRM cycle ergometer (Ingenieurburo Schoberer, Julich, Germany). Power output was sampled at 1-s intervals. The SRM crank system has been demonstrated as valid and reliable in determining maximal and submaximal power output data (18). Expired gases were continually monitored through a mass spectrometer (EX670, Morgan Medical Ltd., Gillingham, UK) for determination of O_2. Before each test, the system was calibrated with known volumes (3-L Hans Rudolf syringe, Hans Rudolph, Kansas City, MO) and concentrations of gas (O_2, CO_2, N_2, Ar). Heart rate was recorded every second during the test, using a portable heart rate monitor (Polar Vantage, Finland). Finger-prick blood samples were analyzed for whole-blood lactate, using a portable lactate analyzer (Biosen 5030, EKF, Industrie, Elektronic, GmbH, Barleben, Germany). The reliability and validity of this device have previously been determined (technical error of measurement = 1.4% (11)).

Belief manipulation. Before the performance trials, and with the aim of catalyzing or reinforcing beliefs about caffeine, subjects were given a 90-min oral presentation and provided with literature reviewing both the findings of published research into caffeine and cycling performance, and detailing anecdotal evidence relating to the use of caffeine among elite cyclists. The efficacy of this manipulation of beliefs was supported by data collected in poststudy interviews.

Preexperimental protocol. Subjects visited the laboratory on several occasions before commencing the study, to undertake pilot and familiarization trials. On the first visit, each participant performed an incremental $\dot{V}O_{2max}$ test on the SRM cycle ergometer. The SRM was set up to replicate exactly the subjects' habitual riding position. During the two remaining preexperimental sessions, subjects performed 40-km time trials designed to habituate them to the protocol and minimize subsequent learning effects.

Substance administration. The validity of the balanced placebo design hinges on the credibility of the experimental deception. The antiplacebo condition—that is, informed no treatment/received caffeine—presents a significant procedural challenge, specifically, the requirement that an active substance be administered without the subject's awareness. Before data collection, extensive pilot work with a different group of subjects was carried out to assess a variety of methods for the administration of caffeine in this manner. The most effective method, and that chosen for the present study, was to require subjects to consume 200 mL of chilled saline at the start of the warm-up, ostensibly to aid voluntary hydration and to maintain electrolyte balance during the time trials. The caffeine dose (5 $mg \cdot kg^{-1}$ of caffeine anhydrous; Gee Lawson, UK) was thus administered to subjects in chilled saline, and at no point during the study was caffeine administered via capsule. Thus, in the antiplacebo condition, before which subjects were explicitly informed that they were receiving no treatment, no capsule was administered, whereas in the caffeine condition, subjects were administered a placebo capsule (200 mg of cornflour, Sainsbury's, UK, in a gelatine capsule) purporting to contain the caffeine to maintain the belief that the substance was being administered in this manner. An association between saline and a bitter taste was encouraged through administration of caffeine in saline in the caffeine condition, and this was reinforced in the subsequent antiplacebo condition. In placebo (i.e., informed caffeine/received placebo), no treatment (i.e., informed no treatment/received no treatment), and baseline conditions, pure saline was administered. Thus, the order of experimental conditions—caffeine, antiplacebo, placebo, and no treatment—was set to minimize changes in the taste of the saline. Counterbalancing of experimental conditions was, thus, not possible, because switching between conditions and, thus, between the pure and caffeinated saline, might produce such divergent gustatory cues as to potentially alert subjects to their treatment assignment.

Experimental protocol. During a period of 14 wk, subjects performed 14 × 40-km time trials, each preceded by a standardized, progressive 20-min warm-up. Initial pilot work demonstrated that the CV for power for this protocol was 1.5% (95% confidence interval (CI) 1.0 to 3.1). Eight experimental trials were interspersed with six baseline trials (Fig. 1). Multiple baseline trials were incorporated to provide a measure of change over the study that might be attributable to order or training effects. As is depicted in Figure 1, treatment effects were estimated in relation to adjacent baselines; that is, the mean of the two trials per

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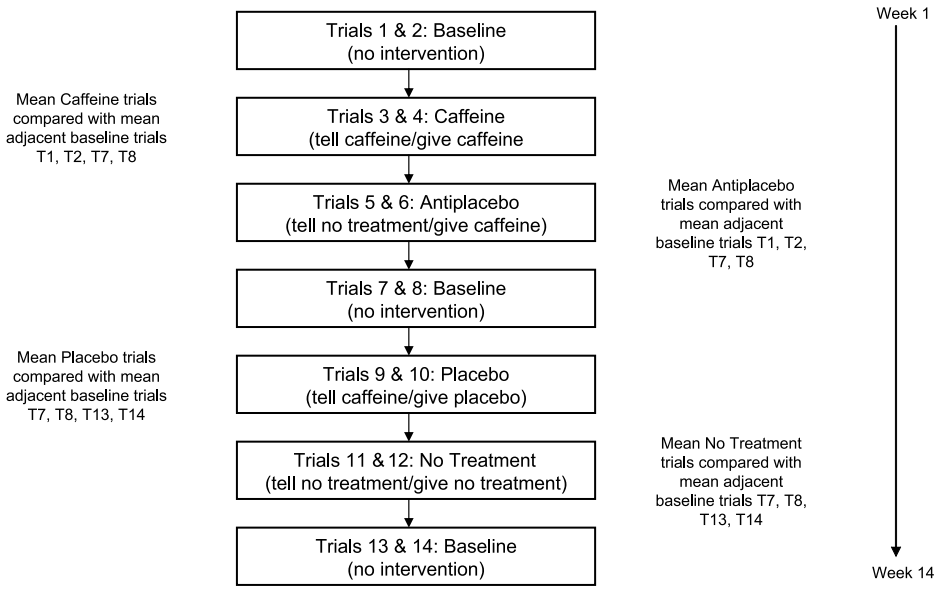


FIGURE 1—Distribution of baseline and experimental trials showing experimental–baseline comparisons.

experimental condition was compared with the mean of the most temporally proximal baseline trials. Trials commenced from a rolling start (100 rpm), and subjects had free choice over gears, cadence, and movement on and out of the saddle. Subjects were instructed to complete the trials in the shortest time possible. Measures were mean power (W); $\dot{V}O_{2max}$ ($mL \cdot kg^{-1} \cdot min^{-1}$), determined as the highest consecutive breath-by-breath $\dot{V}O_2$ point obtained during any 60-s period of the test; mean heart rate (bpm); and mean blood lactate (mM), taken at the thumb every 10 km and 3 min after testing. Ratings of perceived exertion (8) were not used, because these may have drawn the subjects' attention to sensory and perceptual cues during performance, thereby increasing the risk of unblinding. To limit the potential for subjects to employ any pacing strategies based on, for example, knowledge of performance in previous trials, subjects received only information pertaining to elapsed distance. No data relating to time, power, or heart rate were available. No encouragement was given to subjects during the trials, and all performance feedback (power, time, speed, cadence, heart rate, $\dot{V}O_2$, and lactate) was reserved until the end of the study. Environmental conditions were standardized during all tests; temperature and relative humidity were recorded before each trial using a sensormeter (Philip Harris, Shenstone, England) and were maintained at 20–21°C and 50–60%, respectively. Barometric pressure was recorded via a barometer (F. D. and Co. Ltd, Watford, England). A constant fan speed was provided during exercise, and the same subject-selected hydration protocol was used during each test.

Performance trials were undertaken on an individual basis and were usually conducted once per week per individual. Each individual subject's trials were conducted at the same time of day to control for any systematic diurnal variation in performance. Subjects were instructed to abstain from all caffeine-containing foods and beverages for 24 h before testing, to control for possible effects of caffeine already consumed, and they were asked not to use any other potentially ergogenic substances such as creatine monohydrate, sodium bicarbonate, and glycerol during the course of the study. Subjects were requested to otherwise prepare for the trials as they would for an athletic competition (i.e., fully hydrated, well rested, consuming a high-carbohydrate diet) and to repeat the same training and food selection for subsequent tests. Before baseline trials, subjects were informed, "Today you are performing a baseline trial." Before informed caffeine/received caffeine and informed caffeine/received placebo trials, subjects were instructed, "Today you will be given caffeine," and before informed no treatment/received caffeine and informed no treatment/received no treatment, subjects were instructed, "Today you are receiving no treatment" (note that *received placebo* refers to a placebo intervention—that is, the administration of a capsule containing an inert substance—whereas *receive no treatment* describes a condition in which no intervention is administered).

Statistical analyses. All data were tested for homogeneity of variance and normal distribution. Repeated-measures ANOVA were run to identify differences between baseline power and, thus, to identify any training or learning effects. Difference scores for power relative to baseline were calculated for each condition, and a 2 (drug: received caffeine vs received no caffeine) × 2 (belief: received caffeine vs received no caffeine) factorial ANOVA was conducted to identify main effects for belief and caffeine, and any interactions. Repeated-measures ANOVA was used to identify differences between baseline and experimental conditions for mean power, $\dot{V}O_{2max}$, and mean

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lactate. The Greenhouse–Geisser epsilon was reported when the sphericity assumption was violated. Nonparametric heart rate data were analyzed using the Friedman test. Data are expressed as means ± SD. Maximal values for heart rate and blood lactate per trial were not analyzed. Where practical, precision of estimates of outcome statistics are reported as 95% confidence limits of the difference between conditions, and as probabilities that the true effect is beneficial, trivial, and/or harmful (4).

In relation to the balanced placebo design, arguments have been made both for and against excluding from analyses those subjects who failed to believe the instructional set (9,23). For the present study, rates of deceived and undeceived subjects were derived from responses to posttest interview questions regarding the credibility of the experimental manipulation. Delineation of subjects by blinding status enabled analyses of both the subsample of 12 subjects who were successfully deceived and the full sample of 14 subjects to be conducted. Findings from the two analyses were nearly identical, perhaps reflecting that whereas two subjects voiced doubts as to the purported nature of the investigation, they were unable to identify any specific occasion where they felt that they had been deceived. Although the inclusion of potentially undeceived subjects arguably challenges the orthogonal assumption of the balanced placebo design (23), including deceived and excluding undeceived subjects produces biases in subject selection and problems with the generalizability of the results (9). Analyses based on the full sample of 14 subjects are, therefore, reported.

RESULTS

Baseline data. No significant differences between individual baseline trials (baselines 1–6) were evident. However, when grouped as pre (baselines 1 and 2), mid (baselines 3 and 4), and post (baselines 5 and 6), a substantial increase in power was evident between the pre- and midbaselines (8.5 ± 11.3; 95% CI 0.4–16.5), suggesting a trend towards enhanced performance midstudy (Fig. 2). This trend justified the decision to compare experimental conditions and their *adjacent* baselines, as depicted in Figure 1 (a subsequent comparison of each experimental condition with mean baseline 1–6, however, yielded similar findings). Because variations in human performance are reported to be better modeled as percentages of a subject's true performance than as absolute values (15), and because the difference score relative to adjacent baseline performance was considered to provide a more parsimonious representation of the response to treatment, analyses based on percent difference scores between experimental trials and their respective adjacent baseline trials are reported.

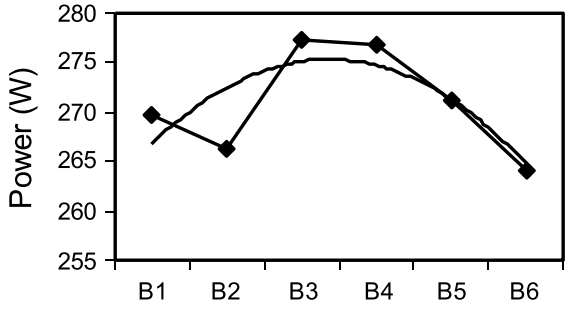


FIGURE 2—Mean power output for six baseline trials and curvilinear trend. B1–B6, baseline trials 1–6.

Experimental data. Results of a 2 (belief: received caffeine vs received no caffeine) × 2 (drug: received caffeine vs received no caffeine) factorial analysis of difference scores in power are presented in Table 1. (Note that belief in caffeine ingestion refers to both conditions in which subjects were informed that they had ingested caffeine—that is, the placebo condition and the caffeine condition.) Results suggest that relative to baseline, caffeine enhanced performance regardless of whether the participant believed that caffeine had been ingested. However, a substantial interaction between belief and pharmacology indicated that this effect was greater in conditions in which subjects were informed that they had not ingested caffeine, whereas in the absence of caffeine, belief exerted a greater influence on performance (2.6%; −0.7 to 5.9; $P = 0.11$).

Results of a repeated-measures ANOVA of difference scores are presented in Table 2. Clear and substantial effects over baseline are associated with conditions in which caffeine was administered. It is also evident that in the absence of ingested caffeine, the belief that caffeine had *not* been ingested resulted in a possibly harmful decrement in performance—that is, a negative placebo or "nocebo" effect (−1.9%; −4.1 to 0.3; $P = 0.08$).

The within-subject CV in the deceptive conditions (antiplacebo and placebo) at 2.8% was 1.7 times larger than the CV when subjects were truthfully informed that they were receiving caffeine (1.7%), indicating the possibility of some disparity between internal sensations and instructions among some subjects. As Hopkins and Wolfinger (16) have noted, whereas a change in mean performance in response to a treatment does not directly affect the estimate of reliability, individual differences in response to the treatment manifest themselves as a decrease in reliability.

Data relating to measured physiological variables by condition are presented in Table 3. Effects over baseline for mean heart rate and mean lactate concentration were clear; no clinically significant differences were observed in mean heart rate, but substantial differences in lactate were observed, the distinction being most evident between the antiplacebo and no treatment conditions (24 ± 31%; −1 to 50%; $P = 0.06$) and the caffeine and no treatment conditions (19 ± 25%; −2 to 40%; $P = 0.08$). Greater lactate was produced after the receipt of caffeine independently of belief. Results for mean $\dot{V}O_{2max}$ were less clear, with possibly clinically significant increases in maximal oxygen

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TABLE 3. Effects of experimental conditions on maximum oxygen uptake, mean heart rate, and mean blood lactate concentration.

	Maximum Oxygen Uptake			Mean Heart Rate			Mean Lactate Concentration		
		Change (%) from Mean Baseline			Change (%) from Mean Baseline			Change (%) from Mean Baseline	
Condition	Mean ± SD ($mL \cdot kg^{-1} \cdot min^{-1}$)	Mean ± SD	95% CI	Mean ± SD (bpm)	Mean ± SD	95% CI	Mean ± SD (mM)	Mean ± SD	95% CI
Inform no treatment/ give no treatment	51.1 ± 5.5	−0.1 ± 6.8	−4.1 to 3.8	155 ± 10	0.2 ± 2.5	−1.2 to 1.6	5.6 ± 1.6	−8 ± 26	−23 to 7
Inform caffeine/ give caffeine	52.5 ± 3.7	1.8 ± 5.1	−1.1 to 4.7	158 ± 9	1.1 ± 2.4	−0.3 to 2.5	6.6 ± 1.7	11 ± 29	−6 to 28
Inform no treatment/ give caffeine	52.8 ± 4.7	2.4 ± 8.6	−2.6 to 7.3	156 ± 8	0.2 ± 3.1	−1.6 to 2.0	6.9 ± 2.0	16 ± 35	−4 to 36
Inform caffeine/ give placebo	52.0 ± 4.4	1.8 ± 7.9	−2.8 to 6.4	155 ± 8	0.3 ± 1.7	−0.7 to 1.3	6.2 ± 1.8	2 ± 27	−14 to 18

95% CI, 95% confidence interval.

uptake observed in all conditions except for informed no treatment/received no treatment.

Within-subject CV were calculated at 3.0% for baseline, and 2.7% for experimental power—values reasonably comparable with previous research. For example, Smith et al. (26) have reported CV for power of 2.5% for 40-km field time trials and 2.0% for 40-km time trials in laboratory conditions, using a caliber of athlete similar to that used in the present study. Whereas the CV suggests that the observed performance effects are unlikely to be the results of random biological or mechanical variation, use of multiple baselines enabled any variation to be identified and factored into the analysis.

DISCUSSION

Collectively, our results indicate that the ingestion of caffeine was associated with a substantial increase in power over baseline in 40-km laboratory cycling performance. On the basis of Batterham and Hopkins' (4) probabilities of practical and clinical significance, such an improvement is highly likely to be worthwhile to a competitive cyclist. This finding supports the widely reported proposal that caffeine is a useful ergogenic aid in endurance performance. Using the same criterion, the main effect of belief was likely trivial, and no placebo effect in the form of a substantial increase in power in the informed caffeine/received placebo condition was observed. A potentially harmful *negative* effect on performance was, however, associated with the informed no treatment/received no treatment condition. A substantial interaction between pharmacology and belief indicated, somewhat surprisingly, that the effect of caffeine was greater when subjects believed they had *not* ingested caffeine than when they believed they had. Similarly, the combined placebo/nocebo effect observed in conditions in which caffeine was not present was substantially greater than that observed in conditions in which it was.

Given the aims and methods of the study, perhaps the key finding above is that a clear, positive placebo effect was not observed. This runs counter to the findings of previous sport research of substantially improved performance in conditions in which subjects believed they had ingested an ergogenic substance (2,6,7,10,21). Certainly, given that subjects produced greater power in the informed caffeine/received placebo condition than in the informed no treatment/received no treatment condition, a placebo effect could be inferred. However, performance in the former condition was only marginally better than at baseline, suggesting that in the absence of caffeine, the negative effect of negative belief on performance was somewhat more substantial than the positive effect of positive belief. It has been reported elsewhere that experimental subjects might perform at a level *below* baseline when they believe either that they have received no treatment or that the experimental intervention is likely to have a negative effect (6,7). Such nocebo effects have received little attention in the sports science literature and warrant further investigation.

The finding that caffeine exerted a greater effect in trials in which subjects were falsely informed they had *not* ingested it than in trials in which they were correctly informed they had is somewhat counterintuitive. That is, conventional wisdom would suggest that the belief of caffeine ingestion would augment its ergogenic effect, and vice versa. It is possible that this observed effect was the result of sampling error, or possibly of expectancy or conditioning effects (see below). Also, given that Beedie et al. (7) have reported that subjects adopted different pacing strategies depending on whether they believed they had or had not ingested caffeine, the observed effect might have resulted from subjects producing less volitional power in trials in which they believed caffeine would augment their performance. However, this proposal is undermined by the fact that this apparent positive effect of negative belief on performance in the presence of caffeine was *reversed* in the absence of caffeine.

Given that the belief effects reported above seem more substantial in the absence of caffeine, and that previous research (7) has reported apparently unambiguous placebo effects of caffeine on cycling performance also observed in the absence of caffeine, it is not unreasonable to speculate that placebo/nocebo effects might operate somewhat differently depending on whether an active substance is present. It is interesting to note that of the studies that have reported placebo effects in sports performance (2,6,7,10,21), only in the study by Clark et al. (10) was an active substance

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actually administered; subjects in the remainder were simply led to *believe* they had received an active substance. Thus, little research in sport has addressed the placebo effect associated with a substance in the presence of that substance, which is also the case with a significant percentage of placebo effect research elsewhere. It is not unreasonable to suggest that when subjects receive an active substance, they are consciously or unconsciously able to respond to the often-subtle cues that might suggest its presence. This process might also allow comparison of subjective perceptions in experimental conditions in which the substance is present and those in which it is not. Therefore, in a study in which no active substance is administered, no comparison of the presence or otherwise of that substance is made, and the subject's belief about the intervention might be *the* significant factor in bringing about an enhancement in performance. However, in a design in which both placebo and active substances are administered, the subject is in a position to make such a comparison, even at a level below conscious awareness, and it might be this comparison, as opposed to the experimental instruction, that is responsible for any enhancement in performance. Although pilot work before the present study suggested that the experimental blind (chilled saline solution) was effective, it is possible that a degree of habituation to the effects of caffeine on performance in the habituation and early experimental trials facilitated the type of comparison described above in the subsequent trials in which caffeine was *not* administered. Such a process would have subverted the blind, logically resulting in the trivial placebo effect in the informed caffeine/received placebo condition observed above. We acknowledge that the failure to counterbalance conditions in the present study is, thus, a limitation. However, this strategy was adopted to preclude the possibility that subjects would detect the presence of caffeine after trials in which they had received a placebo. A between-subjects design might have allowed the authors to overcome these problems, although such a design might well have presented problems relating to variability in placebo and caffeine responsiveness between groups (see below), rendering comparisons unreliable. To return to the original point above, it is not unreasonable to speculate that, all other things being equal, the findings of a placebo effect study in which an active substance is administered and in which beliefs are also manipulated might be somewhat different from those of a study in which only the beliefs are manipulated and no active substance is administered; future researchers should consider this issue.

An alternative explanation for the findings above is that, given that the anticipated improvement in performance in the informed caffeine/received placebo condition did not materialize, it is possible that the sample in the present study was not representative of a normal athletic population in terms of placebo responsiveness. Data from previous research in sport suggest that placebo responsiveness varies considerably between individuals. For example, although it has been reported that approximately 30% of 42 team-sport athletes responded to a placebo intervention (6), a rate that is consistent with Beecher's often-cited proposal that approximately one third of the population is placebo responsive (5), another study has reported that approximately 70% of seven competitive cyclists responded to a placebo. Interestingly, Beedie et al. (7) as well as Clark et al. (10), in reporting studies that examined laboratory cycling performance, suggest that low or zero placebo responding was associated with higher levels of performance, and on this basis it is an intriguing possibility that placebo responsiveness might be a function of the status or the experience of the athlete in question. Such a proposal is supportable from many standpoints. It is logical to suggest that the higher the status of the athlete, the more likely it is that he or she would be better able to attain or maintain an optimal level of performance, thus leaving less, or even zero, "headroom" between that level of performance and the athlete's optimal or ceiling level. Arguably, it is in this gap between present and optimal levels of performance that any placebo effect can operate. If an athlete is performing optimally, it is logical to argue that he or she will not, or cannot, respond positively to a placebo, at least in the form of enhanced performance, unless, of course, that placebo is able to produce an instantaneous increase in, for example, lactate tolerance or oxygen uptake—effects that seem highly unlikely and that certainly have not been suggested in any previous research. Equally, it is possible that placebo responsiveness might in itself represent a deficit in an athlete's psychological makeup, thereby preventing that athlete from attaining consistent levels of optimal performance. Support for this proposal has been presented by Beedie et al. (6), who report that, over and above well-documented positive placebo effects, negative information about an intervention exerted a negative effect on performance. It is certainly possible that, given the relatively high mean age and experience of the subjects in the present study (43 and 21 yr, respectively), the observed lack of a placebo response in the informed caffeine/received placebo condition was itself a function of the sample.

A final possible explanation for the findings above relates to the issue of expectancy. Subjects were instructed either that they *definitively had* or *definitely had not* received caffeine before each trial. Much previous research into the placebo effect has left subjects unsure as to whether they would receive caffeine or a placebo (as is, of course, the case in the traditional placebo-controlled design). It has often been suggested in the literature that this uncertainty about treatment allocation reduces the magnitude of observed effects in comparison with conditions in which subjects have 100% certainty (20). Recent research has, however, suggested that a degree of uncertainty might, in fact, be required to elicit a placebo effect (12,14). For example, Fiorillo et al. (14) have demonstrated that placebo-induced dopamine activation is maximal when the probability of experiencing a beneficial outcome is 0.5. This

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is a somewhat counterintuitive idea, but it may lend support, in sports performance research at least, to the idea that the placebo effect might be more of a factor in laboratory research in which subjects are given a 50% expectation of treatment effect than in the real world (or, indeed, in the present study), when one would imagine that an athlete has a 100% expectation of a treatment effect. In fact, this proposal would suggest that the placebo effect in such research is in fact a *consequence* of the very design employed to control for it! From a psychological perspective, it is possible that the emotion of hope, usually elicited in situations in which the likelihood of a desired positive outcome occurring is significantly less than 100%, might be a significant factor in eliciting a placebo response. Such an argument would make sense from a psychoevolutionary standpoint and is worthy of further research.

It is often suggested that the placebo effect is a factor in sports performance, especially in relation to the use of ergogenic aids. However, the idea that such belief effects are a constant, predictable presence in such settings is questioned by the findings of the present study. In fact, the substantive finding above was that caffeine works whether subjects believe they have ingested it or not, whereas arguably the most surprising finding was that a placebo effect, which in the present study would have been manifest as a substantial improvement in performance in the informed caffeine/received placebo condition, was not observed. Certainly, the findings above may be anomalous; it has, for example, been demonstrated that placebo responsiveness varies from sample to sample. However, several key authors in medicine and psychology have questioned the idea of the "ubiquitous placebo." These authors have not necessarily suggested that the placebo effect does not exist, but they have indicated that it may be far more complex and unpredictable than is often suggested. In keeping with such suggestions, the data above indicate that belief and pharmacology not only interact, but that they might do so in a quite complex and counterintuitive manner, with potential implications for both research and applied settings. Future research should investigate the placebo and nocebo effect further, identifying psychological and environmental factors that might influence the magnitude and direction of this rather mercurial phenomenon in relation to sports performance.

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